

RAG Indexing, Document Chunking, and Semantic Fidelity

How Source Knowledge Changes Before Retrieval Even Begins

AI Memory and Reality Drift - Note #2 v1

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The Basic Pattern

A RAG system does not retrieve directly from a source document. Before retrieval, the source is parsed, cleaned, segmented, annotated, embedded, and stored in an index. These operations determine what the system will later be capable of finding.

Chunking is especially consequential. A document written as one connected argument becomes a set of retrieval units. Definitions may be separated from qualifications. A conclusion may be detached from the evidence supporting it. A policy rule may be retrieved without the exception appearing on the next page.

The retrieval system can perform perfectly against the index while the index no longer preserves the structure that made the source meaningful.

When the Index Becomes the Source

Indexing pipelines are designed for searchability. Fixed-size chunks simplify processing. Semantic chunks attempt to preserve topical boundaries. Parent-child systems retrieve small passages while restoring larger context. Each approach chooses a different tradeoff between precision, coverage, and continuity.

Once the original documents are hidden behind an application, the indexed representation becomes the model's effective source environment. Retrieval quality is then measured against what the index contains, not what the documents originally expressed.

How Pre-Retrieval Drift Emerges

Stage 1 - Parsing. Files are converted into machine-readable text, sometimes losing layout, tables, footnotes, or reading order.

Stage 2 - Segmentation. Text is divided into retrieval units according to token counts, headings, topics, or propositions.

Stage 3 - Enrichment. Metadata, summaries, inferred titles, and contextual labels are attached.

Stage 4 - Embedding. Each unit is compressed into a vector representation optimized for retrieval.

Stage 5 - Index authority. The transformed corpus becomes the world against which later retrieval succeeds or fails.

Semantic Fidelity and Reality Drift

Semantic Fidelity asks what meaning survives transformation. In a RAG index, lexical content may survive while referential, causal, temporal, or pragmatic relationships are weakened. A chunk can quote the correct sentence yet lose the conditions under which it is true.

Reality Drift emerges when the system remains searchable and produces plausible answers while its retrieval substrate becomes progressively less answerable to the original source structure. The failure appears downstream, but its cause may have entered before the first query was issued.

The Representation Stack

Source environment becomes document. Document becomes parsed text. Parsed text becomes chunk. Chunk becomes enriched record. Record becomes embedding. Embedding becomes retrieved context. Retrieved context becomes answer.

Each layer improves machine usability while increasing distance from the source. The architecture remains grounded only when later representations can be decomposed back to intact source passages and their surrounding context.

Examples Across AI Systems

Legal RAG: A rule can be separated from a limiting clause, jurisdiction, or later amendment.

Scientific RAG: A result can be detached from methods, sample limits, uncertainty, or a negative comparison.

Enterprise RAG: A procedure can be retrieved without the version, department, or exception that controls its use.

Multimodal documents: Tables, diagrams, captions, and page structure can disappear during text extraction.

Conclusion

RAG indexing is not neutral preparation. It is the first interpretation of the knowledge base. The pipeline decides which boundaries and relations will remain available to retrieval.

A system cannot recover context that its memory architecture never preserved. Grounding begins at ingestion, not generation.

Related Phrases and Concepts

- RAG indexing
- document ingestion
- document chunking
- semantic chunking
- retrieval units
- context fragmentation
- metadata enrichment
- parent-child retrieval

Semantic Fidelity Project Resources

- [Research Library \(GitHub\)](#)
- [Articles & Essays \(Substack\)](#)
- [Semantic Fidelity Glossary](#)